

Science 24 – Formula Sheet

Formulas

$$\text{Watts} = \frac{\text{Joules}}{\text{second}}$$

$$\text{Energy} = \text{Power} \times \text{Time}$$

$$\text{Kilowatt hours} = \text{kiloWatts} \times \text{hours}$$

$$\% \text{ Efficiency} = \frac{\text{useful energy output}}{\text{total energy input}} \times 100$$

$$\text{Velocity} = \frac{\text{distance}}{\text{time}}$$

OR

$$\text{Slope} = \frac{\text{rise}}{\text{run}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$\text{Momentum} = \text{mass} \times \text{velocity}$$

OR

$$\text{Impulse} = \text{force} \times \text{time}$$

$$\text{Force} = \frac{\text{mass} \times \text{change in velocity}}{\text{time}}$$

Conversions

$$1 \text{ km} = 1000 \text{ m}$$

$$1 \text{ kW} = 1000 \text{ W}$$

$$1 \text{ kg} = 1000 \text{ g}$$

$$1 \text{ Hour} = 3600 \text{ seconds}$$

$$\text{Distance} = \text{Velocity} \times \text{Time}$$

$$\text{Time} = \frac{\text{distance}}{\text{velocity}}$$

Co-ordinates are always (x,y)

$$\text{Velocity} = \frac{\text{momentum}}{\text{mass}}$$

metal

metalloid

non-metal

natural

synthetic

C

Br

H₂O

Np

solid

liquid

gas

synthetic

atomic number

symbol

name

atomic mass

molar mass

(g/mol)

8

O

oxygen

16.00

2-

(if more than one, first one is the most common)

ion charge (if more than one, first one is the most common)																																																																																																																																																																																																																																																																																																																																																																																																																																																		
1		2		3		4		5		6		7		8		9		10		11		12		13		14		15		16		17		18																																																																																																																																																																																																																																																																																																																																																																																																																
1	1⁺ 1⁻ hydrogen 1.01	2	1⁺ 1⁻ 0 oxygen 16.00	3	1⁺ 1⁻ lithium 6.94	4	2⁺ beryllium 9.01	5	1⁺ 1⁻ sodium 22.99	6	1⁺ 1⁻ lithium 6.94	7	2⁺ magnesium 24.31	8	1⁺ 1⁻ potassium 39.10	9	2⁺ calcium 40.08	10	1⁺ 1⁻ rubidium 85.47	11	2⁺ strontium 87.62	12	1⁺ 1⁻ cesium 132.91	13	2⁺ barium 137.33	14	1⁺ 1⁻ francium (223)	15	2⁺ radium (226)	16	1⁺ 1⁻ potassium 39.10	17	2⁺ calcium 40.08	18	1⁺ 1⁻ rubidium 85.47	19	2⁺ strontium 87.62	20	1⁺ 1⁻ cesium 132.91	21	2⁺ barium 137.33	22	1⁺ 1⁻ francium (223)	23	2⁺ radium (226)	24	1⁺ 1⁻ potassium 39.10	25	2⁺ calcium 40.08	26	1⁺ 1⁻ rubidium 85.47	27	2⁺ strontium 87.62	28	1⁺ 1⁻ cesium 132.91	29	2⁺ barium 137.33	30	1⁺ 1⁻ francium (223)	31	2⁺ radium (226)	32	1⁺ 1⁻ potassium 39.10	33	2⁺ calcium 40.08	34	1⁺ 1⁻ rubidium 85.47	35	2⁺ strontium 87.62	36	1⁺ 1⁻ cesium 132.91	37	2⁺ barium 137.33	38	1⁺ 1⁻ francium (223)	39	2⁺ radium (226)	40	1⁺ 1⁻ potassium 39.10	41	2⁺ calcium 40.08	42	1⁺ 1⁻ rubidium 85.47	43	2⁺ strontium 87.62	44	1⁺ 1⁻ cesium 132.91	45	2⁺ barium 137.33	46	1⁺ 1⁻ francium (223)	47	2⁺ radium (226)	48	1⁺ 1⁻ potassium 39.10	49	2⁺ calcium 40.08	50	1⁺ 1⁻ rubidium 85.47	51	2⁺ strontium 87.62	52	1⁺ 1⁻ cesium 132.91	53	2⁺ barium 137.33	54	1⁺ 1⁻ francium (223)	55	2⁺ radium (226)	56	1⁺ 1⁻ potassium 39.10	57	2⁺ calcium 40.08	58	1⁺ 1⁻ rubidium 85.47	59	2⁺ strontium 87.62	60	1⁺ 1⁻ cesium 132.91	61	2⁺ barium 137.33	62	1⁺ 1⁻ francium (223)	63	2⁺ radium (226)	64	1⁺ 1⁻ potassium 39.10	65	2⁺ calcium 40.08	66	1⁺ 1⁻ rubidium 85.47	67	2⁺ strontium 87.62	68	1⁺ 1⁻ cesium 132.91	69	2⁺ barium 137.33	70	1⁺ 1⁻ francium (223)	71	2⁺ radium (226)	72	1⁺ 1⁻ potassium 39.10	73	2⁺ calcium 40.08	74	1⁺ 1⁻ rubidium 85.47	75	2⁺ strontium 87.62	76	1⁺ 1⁻ cesium 132.91	77	2⁺ barium 137.33	78	1⁺ 1⁻ francium (223)	79	2⁺ radium (226)	80	1⁺ 1⁻ potassium 39.10	81	2⁺ calcium 40.08	82	1⁺ 1⁻ rubidium 85.47	83	2⁺ strontium 87.62	84	1⁺ 1⁻ cesium 132.91	85	2⁺ barium 137.33	86	1⁺ 1⁻ francium (223)	87	2⁺ radium (226)	88	1⁺ 1⁻ potassium 39.10	89	2⁺ calcium 40.08	90	1⁺ 1⁻ rubidium 85.47	91	2⁺ strontium 87.62	92	1⁺ 1⁻ cesium 132.91	93	2⁺ barium 137.33	94	1⁺ 1⁻ francium (223)	95	2⁺ radium (226)	96	1⁺ 1⁻ potassium 39.10	97	2⁺ calcium 40.08	98	1⁺ 1⁻ rubidium 85.47	99	2⁺ strontium 87.62	100	1⁺ 1⁻ cesium 132.91	101	2⁺ barium 137.33	102	1⁺ 1⁻ francium (223)	103	2⁺ radium (226)	104	1⁺ 1⁻ potassium 39.10	105	2⁺ calcium 40.08	106	1⁺ 1⁻ rubidium 85.47	107	2⁺ strontium 87.62	108	1⁺ 1⁻ cesium 132.91	109	2⁺ barium 137.33	110	1⁺ 1⁻ francium (223)	111	2⁺ radium (226)	112	1⁺ 1⁻ potassium 39.10	113	2⁺ calcium 40.08	114	1⁺ 1⁻ rubidium 85.47	115	2⁺ strontium 87.62	116	1⁺ 1⁻ cesium 132.91	117	2⁺ barium 137.33	118	1⁺ 1⁻ francium (223)	119	2⁺ radium (226)	120	1⁺ 1⁻ potassium 39.10	121	2⁺ calcium 40.08	122	1⁺ 1⁻ rubidium 85.47	123	2⁺ strontium 87.62	124	1⁺ 1⁻ cesium 132.91	125	2⁺ barium 137.33	126	1⁺ 1⁻ francium (223)	127	2⁺ radium (226)	128	1⁺ 1⁻ potassium 39.10	129	2⁺ calcium 40.08	130	1⁺ 1⁻ rubidium 85.47	131	2⁺ strontium 87.62	132	1⁺ 1⁻ cesium 132.91	133	2⁺ barium 137.33	134	1⁺ 1⁻ francium (223)	135	2⁺ radium (226)	136	1⁺ 1⁻ potassium 39.10	137	2⁺ calcium 40.08	138	1⁺ 1⁻ rubidium 85.47	139	2⁺ strontium 87.62	140	1⁺ 1⁻ cesium 132.91	141	2⁺ barium 137.33	142	1⁺ 1⁻ francium (223)	143	2⁺ radium (226)	144	1⁺ 1⁻ potassium 39.10	145	2⁺ calcium 40.08	146	1⁺ 1⁻ rubidium 85.47	147	2⁺ strontium 87.62	148	1⁺ 1⁻ cesium 132.91	149	2⁺ barium 137.33	150	1⁺ 1⁻ francium (223)	151	2⁺ radium (226)	152	1⁺ 1⁻ potassium 39.10	153	2⁺ calcium 40.08	154	1⁺ 1⁻ rubidium 85.47	155	2⁺ strontium 87.62	156	1⁺ 1⁻ cesium 132.91	157	2⁺ barium 137.33	158	1⁺ 1⁻ francium (223)	159	2⁺ radium (226)	160	1⁺ 1⁻ potassium 39.10	161	2⁺ calcium 40.08	162	1⁺ 1⁻ rubidium 85.47	163	2⁺ strontium 87.62	164	1⁺ 1⁻ cesium 132.91	165	2⁺ barium 137.33	166	1⁺ 1⁻ francium (223)	167	2⁺ radium (226)	168	1⁺ 1⁻ potassium 39.10	169	2⁺ calcium 40.08	170	1⁺ 1⁻ rubidium 85.47	171	2⁺ strontium 87.62	172	1⁺ 1⁻ cesium 132.91	173	2⁺ barium 137.33	174	1⁺ 1⁻ francium (223)	175	2⁺ radium (226)	176	1⁺ 1⁻ potassium 39.10	177	2⁺ calcium 40.08	178	1⁺ 1⁻ rubidium 85.47	179	2⁺ strontium 87.62	180	1⁺ 1⁻ cesium 132.91	181	2⁺ barium 137.33	182	1⁺ 1⁻ francium (223)	183	2⁺ radium (226)	184	1⁺ 1⁻ potassium 39.10	185	2⁺ calcium 40.08	186	1⁺ 1⁻ rubidium 85.47	187	2⁺ strontium 87.62	188	1⁺ 1⁻ cesium 132.91	189	2⁺ barium 137.33	190	1⁺ 1⁻ francium (223)	191	2⁺ radium (226)	192	1⁺ 1⁻ potassium 39.10	193	2⁺ calcium 40.08	194	1⁺ 1⁻ rubidium 85.47	195	2⁺ strontium 87.62	196	1⁺ 1⁻ cesium 132.91	197	2⁺ barium 137.33	198	1⁺ 1⁻ francium (223)	199	2⁺ radium (226)	200	1⁺ 1⁻ potassium 39.10	201	2⁺ calcium 40.08	202	1⁺ 1⁻ rubidium 85.47	203	2⁺ strontium 87.62	204	1⁺ 1⁻ cesium 132.91	205	2⁺ barium 137.33	206	1⁺ 1⁻ francium (223)	207	2⁺ radium (226)	208	1⁺ 1⁻ potassium 39.10	209	2⁺ calcium 40.08	210	1⁺ 1⁻ rubidium 85.47	211	2⁺ strontium 87.62	212	1⁺ 1⁻ cesium 132.91	213	2⁺ barium 137.33	214	1⁺ 1⁻ francium (223)	215	2⁺ radium (226)	216	1⁺ 1⁻ potassium 39.10	217	2⁺ calcium 40.08	21

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
57 57 La lanthanum 138.91	58 58 Ce cerium 140.12	59 59 Pr praseodymium 140.91	60 60 Nd neodymium 144.24	61 61 Pm promethium (145)	62 62 Sm samarium 150.36	63 63 Eu europium 151.96	64 64 Gd gadolinium 157.25	65 65 Tb terbium 158.93	66 66 Dy dysprosium 162.50	67 67 Ho holmium 164.93	68 68 Er erbium 167.26	69 69 Tm thulium 168.93	70 70 Yb ytterbium 173.04	71 71 Lu lutetium 174.97	72 72 Hf hafnium 178.49	73 73 Ta tantalum 180.95	74 74 W tungsten 183.84
89 89 Ac actinium (227)	90 90 Th thorium 232.04	91 91 Pa protactinium 231.04	92 92 U uranium 238.03	93 93 Np neptunium (237)	94 94 Pu plutonium (244)	95 95 Am americium (243)	96 96 Cm curium (247)	97 97 Bk berkelium (247)	98 98 Cf californium (251)	99 99 Es einsteinium (252)	100 100 Fm fermium (257)	101 101 Md mendelevium (258)	102 102 No nobelium (259)	103 103 Lr lawrencium (262)	104 104 Rf rutherfordium (261)	105 105 Db dubnium (262)	106 106 Sg seaborgium (266)